A ratio-locked radio signature of a universal shift in the electron mass

Justin Pulford[1, *]

[1] *Unaffiliated*

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A universal fractional shift ε in the electron mass at high redshift imprints a correlated signature across radio observables. Because each observable depends on m_e through a different power, the induced fractional shifts stand in fixed ratios that are set entirely by atomic physics and are independent of ε itself: $+2\varepsilon$ for the hydrogen 21 cm hyperfine frequency, $+1\varepsilon$ for radio recombination lines, -1ε for the electron column inferred from a dispersion measure, -1ε for the synchrotron characteristic frequency at fixed magnetic field, and -2ε for Faraday rotation. We derive each weight and state explicitly the one convention-dependent entry: the synchrotron row reads -3ε if the emitting particle's energy rather than its Lorentz factor is held fixed, and the choice must be declared before the row is read. We show that the pattern separates a varying electron mass from a varying fine-structure constant arithmetically, since α enters the 21 cm frequency at fourth order but does not enter the dispersion delay at all. The hydrogen-to-deuterium hyperfine ratio furnishes an internal control: α and m_e cancel from it identically, so it is preserved under any universal shift and broken only by species-dependent ones. Throughout, ε is treated as a free amplitude to be fitted rather than predicted, so the ratio structure is testable independently of any particular mechanism for the shift. Read through the pattern, the tightest existing 21 cm comparison gives $|\varepsilon| < 1.4 \times 10^{-5}$ at 2σ over $1.17 < z < 1.56$, limited by systematics rather than by precision; molecular lines bound it more tightly still. The pattern identifies the shift rather than sizing it. A single band deviating from its assigned weight falsifies the pattern.

I. INTRODUCTION

Whether the fundamental constants vary in space or time has a long observational literature [1, 2]. Limits come from quasar absorption spectra [3, 4], from the cosmic microwave background [5, 6], and from big-bang nucleosynthesis, the Oklo natural reactor, and laboratory clock comparisons. Most of this work has concentrated on the fine-structure constant.

The electron mass has attracted more recent attention. A larger m_e near recombination moves that epoch earlier and shrinks the sound horizon, which reduces the discrepancy between the Hubble constant inferred from the microwave background and from the local distance ladder [6, 7]. Constraints on such models come mainly from the CMB itself, where the effect is partially degenerate with the standard cosmological parameters.

Radio propagation has been used for this purpose before. Dispersion measures of localised fast radio bursts have been used to bound variation in α and in the proton-to-electron mass ratio [8], and the clustering of burst dispersion measures likewise [9]. Those analyses treat a single observable at a time.

The construction used below, in which a quantity reconstructed with laboratory constants is read against something that does not share its dependence, is not new and is not confined to the radio bands. The thermal Sunyaev–Zel'dovich effect carries the Thomson cross-section, $\sigma_T \propto \alpha^2 m_e^{-2}$, so a Comptonization inferred from it inherits a weight in exactly the sense of Eq. (3). Compared against an independent X-ray determination of the same intracluster gas, that weight has been used to bound variation in α over 119 clusters [10], and over 618 for spatial rather than temporal variation [11]. What follows applies the same logic to a different set of observables, chosen so that the weights differ in sign as well as magnitude.

Different radio observables depend on the electron mass through different powers, so a universal shift in m_e does not move them together: it moves them by fixed, unequal, calculable amounts. The purpose of this paper is to set that structure out explicitly and to show that the pattern of relative shifts constitutes a test which does not require knowing the size of the shift.

The argument is arranged as follows. Section II fixes notation and separates the two kinds of observable involved. Section III derives the five weights. Section IV shows how the pattern separates a varying electron mass from a varying fine-structure constant, and examines an isotopic control that is easily over-read. Section V states what the argument assumes. Section VI works out the sensitivity, sets out which rows can be entered today and which cannot, and says where the resulting bound sits beside the best existing one.

II. SETUP

We consider a universal, species-independent fractional shift in the electron mass,

$$m_e \longrightarrow m_e\,(1+\varepsilon), \qquad |\varepsilon| \ll 1, \tag{1}$$

holding all other constants fixed. We make no assumption about what produces Eq. (1) and we do not predict ε; it is a single free amplitude. The content of this paper

* pulfordj420@gmail.com

is the set of *relative* responses, which do not depend on it.

A. What is held fixed, and why the statement is dimensionless

A shift in a dimensionful constant is not by itself physically meaningful: the magnitude and even the sign of an inferred effect can depend on the choice of units, an objection raised explicitly against earlier 21 cm work on varying constants [12–14]. Equation (1) must therefore be read with its reference declared.

We hold the fine-structure constant and the strong scale fixed, so that Eq. (1) is a statement about the dimensionless mass ratio

$$\mu \equiv m_p/m_e, \qquad \delta\mu/\mu = -\varepsilon. \tag{2}$$

Every row below then restates without reference to a unit system. The two line frequencies become dimensionless when referred to the proton mass, $\nu_{\rm hf}/m_p \propto \alpha^4\mu^{-2}$ and $\nu_{\rm RRL}/m_p \propto \alpha^2\mu^{-1}$, and are observed as apparent redshift offsets, which are dimensionless in any case. The dispersion and rotation rows are ratios of a reconstructed quantity to the true one, with the same units above and below. The synchrotron row is dimensionless when referred to the proton cyclotron frequency, $\nu_c/\nu_{c,p} \propto \mu$.

Recast this way the weight vector is unchanged, so nothing in Sec. III depends on the unit convention. We continue to write the weights against m_e because it is the more familiar form, with Eq. (2) understood throughout.

For an observable $\mathcal{O}$ we write its weight as the logarithmic derivative

$$w_{\mathcal{O}} \equiv \frac{\partial \ln \mathcal{O}}{\partial \ln m_e}, \qquad \text{so} \qquad \frac{\delta\mathcal{O}}{\mathcal{O}} = w_{\mathcal{O}}\,\varepsilon + \mathcal{O}(\varepsilon^2). \tag{3}$$

Two classes of observable respond differently, and the distinction matters for what is measurable.

Line frequencies are one class. For a single line a shift in the rest frequency is absorbed into an apparent redshift: an observer measuring $\nu_{\rm obs}$ and assuming the laboratory $\nu_{\rm rest}$ infers $1 + z_{\rm app} = (1 + z)(1 + w\varepsilon)$. One line alone therefore carries no information about ε. Two lines of different weight do, since their inferred redshifts differ by a known factor.

Dispersion and rotation measures are the second class. These are not frequencies but quantities reconstructed from a measured delay or rotation angle using laboratory constants. A shift in m_e along the line of sight biases the reconstruction even when the physical column is unchanged.

The pattern derived below spans both classes, so it is not degenerate with redshift.

III. THE FIVE WEIGHTS

A. Hydrogen 21 cm hyperfine: $w = +2$

The ground-state hyperfine splitting is the Fermi contact interaction between the electron and nuclear moments [15, 16]. Its scale is the Rydberg reduced by the ratio of the nuclear to the Bohr magneton and by one further factor α^2,

$$\nu_{\rm hf} \propto \alpha^2 \underbrace{\frac{\mu_N}{\mu_B}}_{\propto\, m_e/m_p} \underbrace{R_\infty c}_{\propto\, \alpha^2 m_e} \propto g_p\,\alpha^4\,\frac{m_e^2}{m_p}, \tag{4}$$

giving $w = +2$.

B. Radio recombination lines: $w = +1$

Transitions between high principal quantum numbers are pure Rydberg [17], $\nu = R_\infty c\,(n_1^{-2} - n_2^{-2})$ with $R_\infty \propto \alpha^2 m_e$, so

$$\nu_{\rm RRL} \propto m_e, \qquad w = +1. \tag{5}$$

C. Dispersion measure: $w = -1$

The group delay of a signal traversing a cold plasma is [18]

$$t_{\rm DM} = \frac{e^2}{2\pi m_e c}\,\frac{1}{\nu^2}\int n_e\,dl. \tag{6}$$

The measured quantity is $t_{\rm DM}$; the reported dispersion measure is obtained by dividing out [19] the *laboratory* constant $e^2/2\pi m_e^{\rm lab}c$. If the true electron mass along the path is $m_e^{\rm lab}(1+\varepsilon)$, the delay produced by a fixed physical column $\int n_e\,dl$ is smaller by $(1+\varepsilon)^{-1}$, and the inferred column inherits that factor:

$$\mathrm{DM}_{\rm inferred} = \left(\int n_e\,dl\right)(1-\varepsilon) + \mathcal{O}(\varepsilon^2), \qquad w = -1. \tag{7}$$

The bias is in the *reconstruction*, not in the plasma. This row therefore tests the shift against any independent determination of the same electron column.

D. Synchrotron characteristic frequency: $w = -1$ *at fixed field*

The characteristic frequency of synchrotron emission is [20]

$$\nu_c = \frac{3}{2}\gamma^2\,\frac{eB}{2\pi m_e c}. \tag{8}$$

This row depends on a convention, which must be fixed before it is used. Holding the magnetic field and the emitting particle's Lorentz factor fixed gives $w = -1$ from the explicit m_e in Eq. (8). Holding the particle's energy fixed instead lets $\gamma = E/m_e c^2$ float, contributing a further -2, so that

$$w_{\rm sync} = \begin{cases} -1, & \text{fixed } \gamma \text{ (fixed-field reading)}, \\ -3, & \text{fixed } E \text{ (fixed-energy reading)}. \end{cases} \tag{9}$$

We adopt the fixed-γ reading throughout and quote -1. Which reading is physically appropriate depends on how the emitting population is labelled, and a test that does not state its choice is not well posed. We return to this in Sec. V.

E. Faraday rotation: $w = -2$

The rotation measure of a magnetised plasma is [21]

$$\mathrm{RM} \;=\; \frac{e^3}{2\pi m_e^2 c^4} \int n_e B_\parallel \, dl, \qquad w = -2, \tag{10}$$

again a bias in the reconstructed quantity rather than in the medium.

F. The pattern

Collecting Eqs. (4)–(8):

$$\boxed{\begin{aligned} & w_{21} : w_{\rm RRL} : w_{\rm DM} : w_{\rm sync} : w_{\rm RM} \\ & \qquad = +2 : +1 : -1 : -1 : -2 \end{aligned}} \tag{11}$$

Every entry follows from a textbook scaling; none is adjustable. The amplitude ε is common to all five and cancels from every ratio, so Eq. (11) is a prediction about *correlations* that can be tested without knowing the size of the shift.

IV. SEPARATION FROM A VARYING FINE-STRUCTURE CONSTANT

The competing hypothesis is a shift in α rather than m_e. The two are separable because they produce different weight vectors across the same five bands. Working in Gaussian units, where $e^2 = \alpha\hbar c$ and hence $e \propto \alpha^{1/2}$, the same derivations give the α-weights in Table I.

The distinction is qualitative before it is quantitative. Every α-weight is positive: a shift in the fine-structure constant moves all five observables in the same direction. The m_e-weights change sign across the set, raising two observables and lowering three. Measuring any two bands of opposite m_e-weight therefore separates the hypotheses by the sign of the correlation alone.

TABLE I. Response of each observable to a fractional shift in the electron mass and in the fine-structure constant, $w \equiv \partial \ln \mathcal{O} / \partial \ln X$. The two columns are not proportional, so the hypotheses are distinguishable; the sign structure separates them without reference to precision.

observable	w for m_e	w for α
21 cm hyperfine	$+2$	$+4$
radio recombination line	$+1$	$+2$
dispersion measure (inferred)	-1	$+1$
synchrotron ν_c (fixed field)	-1	$+1/2$
Faraday rotation measure	-2	$+3/2$

The reason is visible in the two reconstructed quantities. The dispersion delay of Eq. (6) carries $e^2/m_e \propto \alpha/m_e$, so α and m_e enter with opposite signs; the same holds for the rotation measure, where $e^3/m_e^2 \propto \alpha^{3/2}/m_e^2$. The line frequencies, by contrast, carry both constants with positive powers. No overall rescaling maps one column of Table I onto the other: the ratios w_α/w_{m_e} across the five rows are 2, 2, -1, $-1/2$ and $-3/4$.

A. The hydrogen–deuterium control

The hyperfine frequencies of hydrogen and deuterium both take the form of Eq. (4) with their own nuclear factors. Their ratio is therefore

$$\frac{\nu_{\rm hf}(\mathrm{H})}{\nu_{\rm hf}(\mathrm{D})} \;=\; \frac{C_{\rm H}\, g_p/m_p}{C_{\rm D}\, g_d/m_d}, \tag{12}$$

in which both α and m_e cancel identically, the angular-momentum coefficients C depending only on the nuclear spins. Logarithmic differentiation confirms $\partial \ln(\text{ratio})/\partial \ln m_e = \partial \ln(\text{ratio})/\partial \ln \alpha = 0$.

The ratio is therefore not a discriminant between a varying m_e and a varying α, since both preserve it. What it tests is universality: whether whatever shifts does so identically for the two isotopes. It excludes species-dependent variation, which is a useful null in its own right, and it is easily misread as evidence for one constant over the other.

V. WHAT IS ASSUMED

The weights in Eq. (11) follow from standard atomic and plasma physics, but the test built on them carries assumptions that should be stated.

Universality. The shift is taken to act identically on every electron. A species- or environment-dependent variation would break the pattern, and the hydrogen–deuterium ratio of Sec. IV is the control for this.

Other constants fixed. We vary m_e alone. A simultaneous variation in α or in the proton mass adds its own weights to each row, and the observed pattern would then be a superposition. Because the weight vectors differ between m_e and α, a sufficiently well-measured set of bands can in principle separate the components; with few bands it cannot.

Linearity. Existing bounds on a varying electron mass from big-bang nucleosynthesis and the microwave background place $|\varepsilon|$ at the percent level or below [22, 23]. We work to first order in ε. For $|\varepsilon| \lesssim 10^{-2}$ the quadratic terms enter at the 10^{-4} level, below the precision of any of the measurements considered here.

Constancy along the path. Each observable integrates along a line of sight or over an epoch. The weights apply cleanly if ε is constant over the relevant interval. If it varies, each row picks up its own weighting of the variation and the ratios are modified in a row-dependent way. Comparisons at matched redshift are therefore preferable to comparisons across widely separated epochs.

The synchrotron convention. As noted in Sec. III, this row reads -1 or -3 according to whether the emitting population is labelled by Lorentz factor or by energy. The choice must be fixed by the physical situation before the row is used, and we have adopted the former.

VI. SENSITIVITY

Suppose each band i yields a measured fractional shift d_i with Gaussian uncertainty σ_i, against the prediction $d_i = w_i\varepsilon$. The maximum-likelihood amplitude and its error are

$$\hat{\varepsilon} = \frac{\sum_i w_i d_i/\sigma_i^2}{\sum_i w_i^2/\sigma_i^2}, \qquad \sigma_\varepsilon = \left(\sum_i \frac{w_i^2}{\sigma_i^2}\right)^{-1/2}. \tag{13}$$

The weights enter quadratically, so the high-$|w|$ rows dominate. For all five bands measured at a common fractional precision σ, $\sum_i w_i^2 = 11$ and

$$\sigma_\varepsilon = \sigma/\sqrt{11} \simeq 0.30\,\sigma. \tag{14}$$

In practice the bands will not be measured at comparable precision, and a two-band comparison is the realistic near-term test. Among pairs, the 21 cm and Faraday rows are the extremes of the set and give $\sigma_\varepsilon = \sigma/\sqrt{8} \simeq 0.35\,\sigma$; a pair involving one unit-weight row would give $\sigma_\varepsilon = \sigma/\sqrt{5} \simeq 0.45\,\sigma$, though for the reasons set out immediately below no unit-weight row is presently measurable, so that configuration is not currently available.

A stronger caveat applies to three of the five rows, and it is not a matter of precision but of kind. Equation (13) presumes each band supplies an independent Gaussian σ_i; the recombination-line, synchrotron and dispersion-measure rows do not. Radio recombination frequencies are *computed* from the Rydberg formula and therefore already carry m_e, so an observed line compared against its own computed rest frequency returns a velocity rather than a constraint; that row enters only in differential comparison against a transition of different m_e dependence. Synchrotron carries a formal degeneracy between the field strength and the emitting population, broken in practice by assuming energy equipartition, so its uncertainty is dominated by a modelling choice that is itself degenerate with the quantity being weighed — and the resulting normalisation correction does not have a fixed sign. The dispersion-measure row fails the same presumption, for the reason set out below: a constant ε is absorbed exactly by the fitted dispersion measure, so the row reports the precision of an independent electron-column determination rather than a σ_i of its own.

We therefore treat only the two rows referred to a fixed laboratory rest frequency (21 cm and Faraday rotation) as presently measurable, giving $\sigma_\varepsilon = \sigma/\sqrt{8} \simeq 0.35\,\sigma$, and regard $\sigma_\varepsilon = \sigma/\sqrt{11}$ as an upper bound on what the method could reach were the remaining three rows made measurable, rather than as a forecast. The five-row pattern of Sec. III is unaffected: it is a statement about how a shift propagates, and all five rows carry it. What is reduced is the number of rows that can presently be turned into a measurement.

Pairs of opposite m_e-weight are preferable for a reason independent of their statistical weight. A systematic common to both bands, entering as a shared multiplicative offset, contributes equally to d_{21} and $d_{\rm RM}$ and therefore cancels in the difference that carries the signal, whereas it would be absorbed into $\hat{\varepsilon}$ by a same-sign pair. The sign structure of Table I is thus useful twice: it separates the m_e and α hypotheses, and it suppresses common-mode systematics.

A. Where the rows currently stand

The two presently measurable rows are quoted below with their statistical figures and, separately, the systematic that actually binds each; in both cases the systematic exceeds the statistical precision by roughly half an order of magnitude, and in each it arises from a different physical effect. The dispersion-measure row is set out between them, not as a third constraint but because its exclusion rests on a degeneracy the reader should be able to check rather than on an assertion.

The 21 cm row is constrained by comparing hyperfine absorption against optical metal lines in the same absorber. The tightest such analysis reports $\Delta x/x = -(0.1 \pm 1.3) \times 10^{-6}$ over four systems at $1.17 < z < 1.56$, where $x \equiv g_p\alpha^2/\mu$ [24]. Since $\mu = m_p/m_e$, holding α and the proton moment fixed gives $\delta \ln x = -\delta \ln \mu = \varepsilon$, so that measurement is a measurement of the amplitude itself:

$$\varepsilon = -(0.1 \pm 1.3) \times 10^{-6} \quad (1.17 < z < 1.56), \tag{15}$$

statistical only. An earlier analysis of the same combination quotes a maximum systematic of 6.7×10^{-6} [25], arising because the radio and optical absorption need not sample the same gas along a finite-width sightline — which is why the tighter analysis restricts itself to milliarcsecond-unresolved background sources. Added in quadrature the two give $\varepsilon = -(0.1 \pm 6.8) \times 10^{-6}$, or $|\varepsilon| < 1.4 \times 10^{-5}$ at 2σ. We stress that this is a reinterpretation of a published measurement under the assumptions of Sec. V, not a new one; and that the systematic exceeds the statistical error by a factor of five, which is the condition a ratio test across bands with independent systematics is meant to improve on.

The dispersion-measure row has by far the best raw precision and, for the same reason as the recombination-line row, the least of it is usable. Median DM uncertainties reach $\sim 10^{-5}\,\mathrm{pc\,cm^{-3}}$ in low-frequency monitoring, which against typical pulsar-timing-array dispersion measures of 10–100 $\mathrm{pc\,cm^{-3}}$ is a fractional precision of 10^{-7}–10^{-6}. That precision does not transfer to ε. A pulsar timing model fits an infinite-frequency arrival time and a dispersion measure from $t(\nu) = t_\infty + K\,\mathrm{DM}/\nu^2$, and a universal shift $m_e \to m_e(1+\varepsilon)$ rescales the delay's coefficient while leaving its ν^{-2} shape untouched. The fit therefore absorbs it exactly, returning $\mathrm{DM_{fit}} = (\int n_e\,dl)/(1+\varepsilon)$ with t_∞ unchanged and no timing residual at any frequency coverage. A constant ε is thus degenerate with the fitted dispersion measure, and improving the DM measurement improves $\mathrm{DM_{fit}}$ while saying nothing about the shift. The row constrains ε only in the sense already stated in Sec. III: against an independent determination of the same electron column, whose precision, not the DM precision, sets σ_ε.

That route has been taken. The fast-radio-burst analyses cited in Sec. I compare the observed dispersion measure against the value predicted from $\Omega_b h^2$ and an assumed intergalactic baryon fraction, which is exactly the external column referred to above. Seventeen bursts with supernovae in a runaway-dilaton parametrization give $\Delta\alpha/\alpha$ at the 10^{-2} level [26]; 50 bursts over $0.004 < z < 1.02$ tighten this to $\Delta\alpha/\alpha \simeq 2 \times 10^{-5}$ [8]. One caution about reading those results as mass constraints. The quoted $\Delta\mu/\mu \simeq -1 \times 10^{-5}$ is not independent of the α result but derived from it, through an assumed relation between mass and coupling variation, so it does not constrain ε at fixed α — the case considered here. The corresponding mass-only analysis has not to our knowledge been performed, and its limiting uncertainty would be the baryon fraction rather than the dispersion measures.

What the timing literature does bound is the *variation* of ε. Mis-estimated dispersion produces infinite-frequency arrival-time offsets of order 20 μs for high-DM pulsars, correlated on month timescales [27, 28]; read as a bound on epoch-to-epoch drift this gives $\sim 2 \times 10^{-6}$ at high DM and low frequency, falling to $\sim 3 \times 10^{-4}$ at 1400 MHz and low column. We therefore do not fold the 10^{-7} figure into any forecast, and we do not treat this row as carrying a constant-ε constraint at all.

The Faraday row is set by rotation-measure catalogue precision. The deepest current grid reports a median RM uncertainty of $0.06\,\mathrm{rad\,m^{-2}}$ across 2461 extragalactic sources, with a possible residual systematic of up to $0.3\,\mathrm{rad\,m^{-2}}$ after ionospheric correction [29].

B. Where the amplitude actually stands

The strongest existing bound on ε comes from none of these rows. Methanol absorption in the $z = 0.88582$ lens toward PKS1830$-$211 gives $|\Delta\mu/\mu| \lesssim 4\times10^{-7}$ at 2σ over $0 < z \leq 0.886$ [30]. The methanol transitions carry μ alone — no α, no proton moment — so that limit constrains ε under weaker assumptions than Eq. (15) requires, and it is some thirty-five times tighter. An independent analysis of the same source at three telescopes, using ten transitions, agrees: $\Delta\mu/\mu = (-1.0 \pm 0.8_{\mathrm{stat}} \pm 1.0_{\mathrm{sys}}) \times 10^{-7}$ [31], which is $|\Delta\mu/\mu| < 3.6 \times 10^{-7}$ at 2σ and is likewise dominated by its systematic. The 21 cm and Faraday pair would need a fractional precision of $\sigma \simeq 1.1 \times 10^{-6}$ per band to match it. The two rows that can presently be entered therefore do not improve the limit on the amplitude, and we do not claim that they do.

What they supply instead is the quantity a single molecular comparison cannot. The point is a counting one. A measurement of μ in one transition returns a single number, against which ε is a single free amplitude; the fit is exact and nothing is left over, so every hypothesis with a free amplitude accommodates the result equally well and the measurement discriminates among none of them. Two rows of different weight return two numbers for the same one amplitude, leaving one degree of freedom that the hypotheses do not share — and it is that residual, not the amplitude, that Eq. (13) is a sensitivity calculation for. Because the count does not involve σ, a tighter bound on the amplitude does not diminish it. How the methanol result was arrived at is instructive, because the same effect governs the 21 cm row. A tighter statistical limit from one line pair was set aside in favour of the weaker figure from three transitions whose profiles agree, on the grounds that only those demonstrably sample the same gas. Both of the best available constraints on ε are thus limited by sightline structure rather than by photon noise.

VII. FALSIFICATION

The pattern is falsified by any single band deviating from its assigned weight, once two or more bands are measured along the same sightline or at matched redshift.

Two outcomes distinguish the hypothesis from the alternatives. A pattern consistent with $+2 : +1 : -1 : -1 : -2$ across the five bands supports a universal shift in m_e. A pattern with the same signs but different magnitudes is evidence for a mixed variation, or for a variation in α, and the weight vectors separate the cases. Uncorrelated shifts, or a single anomalous band with the others null, are not evidence for this hypothesis and should be treated as instrumental or astrophysical in origin.

VIII. CONCLUSION

A universal fractional shift in the electron mass produces a correlated signature across radio observables whose relative sizes are fixed by atomic physics. The ratios $+2 : +1 : -1 : -1 : -2$ are independent of the amplitude of the shift, so the structure can be tested without a prediction for that amplitude and without commitment to any mechanism producing it. The pattern differs from that produced by a varying fine-structure constant, and the hydrogen–deuterium hyperfine ratio provides an internal control on universality that both hypotheses pass and a species-dependent variation fails.

Two of the five rows can be entered at present. Read through the weights, the tightest published 21 cm comparison bounds the amplitude at $|\varepsilon| < 1.4 \times 10^{-5}$ over $1.17 < z < 1.56$, with the systematic five times the statistical error — and methanol absorption bounds it some thirty-five times more tightly still. Neither figure is improved here. The case for the pattern is not that it measures ε better, but that a limit on μ is a single number, which cannot say whether the shift lies in the electron mass or in the coupling, nor whether every species shares it. Five weights of two signs answer both questions, and the ordering of the errors points the same way: a ratio taken between bands whose systematics are unrelated is limited by neither alone.

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